

AI PLACEMENT MANAGEMENT SYSTEM

Complete Database / ER Design

Group 12

1. Database Design Objective

The database supports student profiles and resumes, company and job drives, eligibility, applications, interviews, placements, and AI resume/job analysis. All three pairs should use the same database so the complete placement workflow remains integrated.

2. Complete ER Structure

Roles 1:M Users

Users 1:1 StudentProfiles

Users 1:1 CompanyProfiles

StudentProfiles 1:M Resumes

StudentProfiles 1:M Certifications

StudentProfiles 1:M Projects

StudentProfiles M:M Skills through StudentSkills

CompanyProfiles 1:M JobDrives

JobDrives M:M Skills through JobSkills

JobDrives 1:1 EligibilityCriteria

JobDrives 1:M JobEligibleBranches

StudentProfiles 1:M Applications

JobDrives 1:M Applications

Applications 1:M ApplicationStatusHistory

JobDrives 1:M InterviewRounds

Applications 1:M InterviewSchedules

InterviewSchedules 1:1 InterviewResults

StudentProfiles 1:M PlacementResults

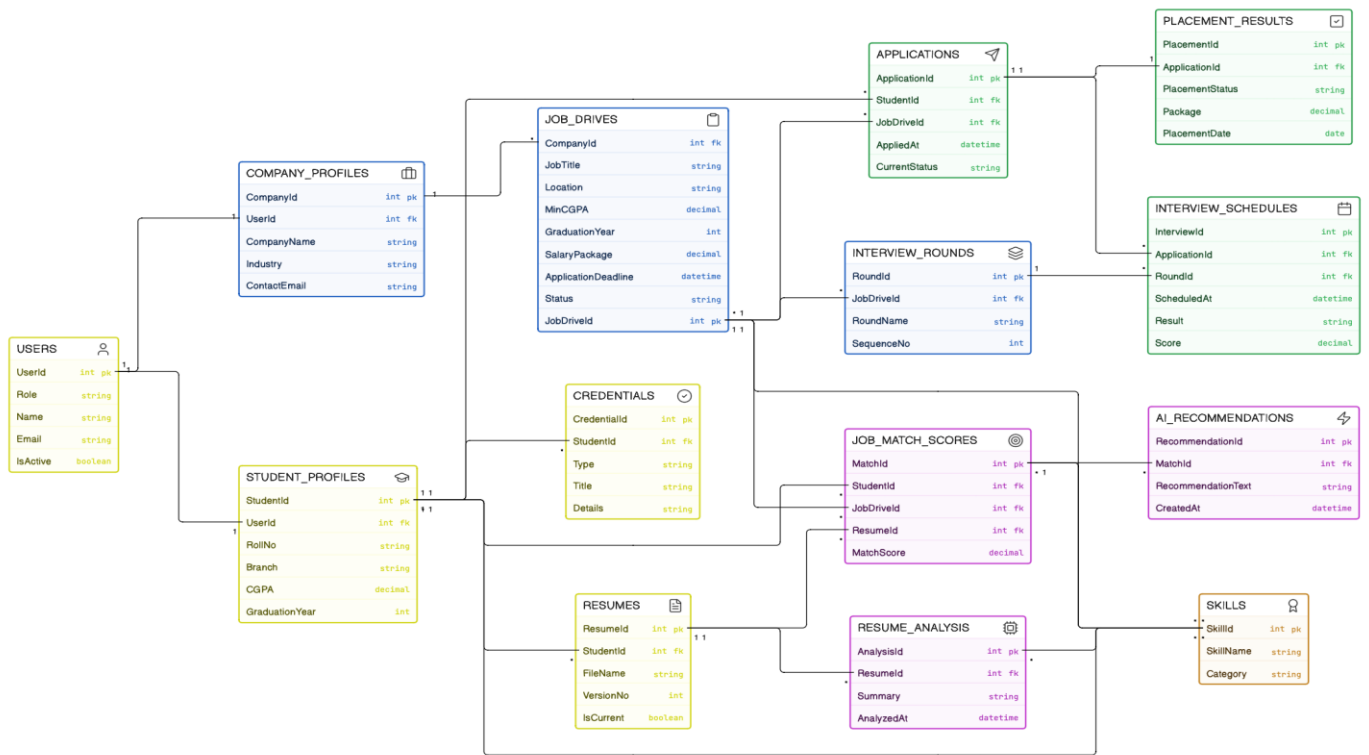
JobDrives 1:M PlacementResults

Resumes 1:M ResumeAnalysis

ResumeAnalysis 1:M ExtractedSkills

JobMatchScores 1:M SkillGaps

Student + JobDrive → JobMatchScores / AIRecommendations



3. Authentication

Roles

Column	Type	Key / Constraint
RoleId	INT	PK
RoleName	VARCHAR(50)	UNIQUE, NOT NULL

Users

Column	Type	Key / Constraint
UserId	INT	PK
RoleId	INT	FK → Roles
Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(150)	UNIQUE, NOT NULL
PasswordHash	VARCHAR(255)	NOT NULL
IsActive	BIT	NOT NULL
CreatedAt	DATETIME	NOT NULL

Suggested roles: Student, Company, Admin.

4. Student Tables

StudentProfiles

Column	Type	Key / Constraint
StudentId	INT	PK
UserId	INT	FK → Users, UNIQUE
RollNo	VARCHAR(30)	UNIQUE
Branch	VARCHAR(100)	
CGPA	DECIMAL(4,2)	
GraduationYear	INT	
Phone	VARCHAR(20)	
DateOfBirth	DATE	

Skills

Column	Type	Key / Constraint
SkillId	INT	PK
SkillName	VARCHAR(100)	UNIQUE
Category	VARCHAR(50)	

StudentSkills

Column	Type	Key / Constraint
StudentSkillId	INT	PK
StudentId	INT	FK → StudentProfiles
SkillId	INT	FK → Skills
ProficiencyLevel	VARCHAR(30)	

Resumes

Column	Type	Key / Constraint
ResumeId	INT	PK
StudentId	INT	FK → StudentProfiles
FileName	VARCHAR(255)	
FilePath	VARCHAR(500)	
UploadedAt	DATETIME	
VersionNo	INT	
IsCurrent	BIT	

Certifications

Column	Type	Key / Constraint
CertificationId	INT	PK
StudentId	INT	FK → StudentProfiles
Name	VARCHAR(150)	
IssuingOrganization	VARCHAR(150)	

IssueDate	DATE	
CredentialUrl	VARCHAR(500)	

Projects

Column	Type	Key / Constraint
ProjectId	INT	PK
StudentId	INT	FK → StudentProfiles
Title	VARCHAR(150)	
Description	TEXT	
Technologies	VARCHAR(500)	
ProjectUrl	VARCHAR(500)	

5. Company and Job Tables

CompanyProfiles

Column	Type	Key / Constraint
CompanyId	INT	PK
UserId	INT	FK → Users, UNIQUE
CompanyName	VARCHAR(150)	
Description	TEXT	
Website	VARCHAR(255)	
Industry	VARCHAR(100)	
ContactEmail	VARCHAR(150)	
ContactPhone	VARCHAR(20)	

JobDrives

Column	Type	Key / Constraint
JobDriveId	INT	PK
CompanyId	INT	FK → CompanyProfiles
JobTitle	VARCHAR(150)	

JobDescription	TEXT	
Location	VARCHAR(150)	
MinCGPA	DECIMAL(4,2)	
GraduationYear	INT	
SalaryPackage	DECIMAL(10,2)	
ApplicationDeadline	DATETIME	
Status	VARCHAR(30)	
ApprovalStatus	VARCHAR(30)	
CreatedAt	DATETIME	

JobSkills

Column	Type	Key / Constraint
JobSkillId	INT	PK
JobDriveId	INT	FK → JobDrives
SkillId	INT	FK → Skills
IsRequired	BIT	

EligibilityCriteria

Column	Type	Key / Constraint
EligibilityId	INT	PK
JobDriveId	INT	FK → JobDrives, UNIQUE
MinCGPA	DECIMAL(4,2)	
MaxBacklogs	INT	
GraduationYear	INT	

JobEligibleBranches

Column	Type	Key / Constraint
JobBranchId	INT	PK
JobDriveId	INT	FK → JobDrives
BranchName	VARCHAR(100)	

JobEligibleBranches is recommended instead of storing multiple branches in one comma-separated field.

6. Applications and Recruitment

Applications

Column	Type	Key / Constraint
ApplicationId	INT	PK
StudentId	INT	FK → StudentProfiles
JobDriveId	INT	FK → JobDrives
AppliedAt	DATETIME	
CurrentStatus	VARCHAR(50)	
RecruiterRemarks	TEXT	

ApplicationStatusHistory

Column	Type	Key / Constraint
HistoryId	INT	PK
ApplicationId	INT	FK → Applications
Status	VARCHAR(50)	
ChangedAt	DATETIME	
ChangedBy	INT	FK → Users
Remarks	TEXT	

InterviewRounds

Column	Type	Key / Constraint
RoundId	INT	PK
JobDriveId	INT	FK → JobDrives
RoundName	VARCHAR(100)	
RoundType	VARCHAR(50)	
SequenceNo	INT	

InterviewSchedules

Column	Type	Key / Constraint
InterviewId	INT	PK
ApplicationId	INT	FK → Applications
RoundId	INT	FK → InterviewRounds
ScheduledAt	DATETIME	
Location	VARCHAR(255)	
MeetingLink	VARCHAR(500)	
Status	VARCHAR(30)	

InterviewResults

Column	Type	Key / Constraint
ResultId	INT	PK
InterviewId	INT	FK → InterviewSchedules, UNIQUE
Result	VARCHAR(50)	
Score	DECIMAL(5,2)	
Remarks	TEXT	
EvaluatedAt	DATETIME	

7. Placement

PlacementResults

Column	Type	Key / Constraint
PlacementId	INT	PK
StudentId	INT	FK → StudentProfiles
JobDriveId	INT	FK → JobDrives
ApplicationId	INT	FK → Applications, UNIQUE
PlacementStatus	VARCHAR(30)	
Package	DECIMAL(10,2)	

PlacementDate	DATE	
OfferDetails	TEXT	

A business rule can restrict a student to one final active placement; confirm this with the team/faculty.

8. AI / ML Tables

ResumeAnalysis

Column	Type	Key / Constraint
AnalysisId	INT	PK
Resumeld	INT	FK → Resumes
AnalyzedAt	DATETIME	
ExtractedText	TEXT	
Summary	TEXT	
ModelVersion	VARCHAR(50)	

ExtractedSkills

Column	Type	Key / Constraint
ExtractedSkillId	INT	PK
AnalysisId	INT	FK → ResumeAnalysis
SkillId	INT	FK → Skills
ConfidenceScore	DECIMAL(5,2)	

JobMatchScores

Column	Type	Key / Constraint
MatchId	INT	PK
StudentId	INT	FK → StudentProfiles
JobDriveId	INT	FK → JobDrives
Resumeld	INT	FK → Resumes
MatchScore	DECIMAL(5,2)	
CalculatedAt	DATETIME	

SkillGaps

Column	Type	Key / Constraint
SkillGapId	INT	PK
MatchId	INT	FK → JobMatchScores
SkillId	INT	FK → Skills
GapLevel	VARCHAR(30)	
Recommendation	TEXT	

AIRecommendations

Column	Type	Key / Constraint
RecommendationId	INT	PK
StudentId	INT	FK → StudentProfiles
JobDriveId	INT	FK → JobDrives
MatchId	INT	FK → JobMatchScores
RecommendationText	TEXT	
CreatedAt	DATETIME	

9. Relationship Summary

Relationship	Cardinality
Role → User	1:M
User → StudentProfile	1:1
User → CompanyProfile	1:1
Student → Resume / Certification / Project	1:M each
Student ↔ Skill	M:M via StudentSkills
Company → JobDrive	1:M
JobDrive ↔ Skill	M:M via JobSkills
JobDrive → EligibilityCriteria	1:1
JobDrive → JobEligibleBranches	1:M
Student → Application	1:M
JobDrive → Application	1:M

Application → StatusHistory	1:M
JobDrive → InterviewRound	1:M
Application → InterviewSchedule	1:M
InterviewSchedule → Result	1:1
Student → PlacementResult	1:M*
JobDrive → PlacementResult	1:M
Resume → ResumeAnalysis	1:M
ResumeAnalysis → ExtractedSkills	1:M
JobMatchScore → SkillGap	1:M

10. Admin and Analytics Data Dependencies

Dashboard Metric	Calculated From
Total Students	StudentProfiles
Total Companies	CompanyProfiles
Total Job Drives	JobDrives
Total Applications	Applications
Selected Students	PlacementResults
Placement Percentage	StudentProfiles + PlacementResults / agreed eligible population
Average Package	AVG(PlacementResults.Package)
Highest Package	MAX(PlacementResults.Package)
Branch-wise Placements	StudentProfiles.Branch + PlacementResults
Company/Job Applications	Applications + JobDrives + CompanyProfiles